

In-class Assignment 6 (Individual): Introduction to GUI, Images, and the Database

AD / SPRINT 3 / 2 — Fall 2026

Note: The in-class assignments are compulsory for completing both the sprint implementation and the overall project. Failure to complete the in-class assignments will result in losing the implementation part of your sprint tasks and, consequently, your overall course grade. **This will cause you to fail the course if your total points do not reach the required threshold of 2 points.**

This assignment requires that you attend the lecture and complete the class exercise at the same time as the teacher.

Convert your Temperature Converter into a JavaFX application. Create a database with at least two related tables to keep track of the speed, distance, and time. The interface must interact with the database.

Tasks

1. Install the XMing application.
2. Add unit tests for your classes (e.g., DBConnection, Main, TempCalculator, TempRecordDAO, TemperatureUnit, TemperatureUnitDAO, TempRecord, etc.).
3. Set up JaCoCo for code coverage.
4. Create a Jenkinsfile.
5. Create a Dockerfile, and run it both locally and through the Jenkinsfile, deploying the image to hub.docker.com.
6. Run the application both locally and as a Docker image.

Submission

1. GitHub repository link
2. URL of the image on hub.docker.com
3. Screenshots of the image running in XMing (or any other X server application)